DBMS ASSIGNMENT

ASSIGNMENT 6

Implementation of Data Integrity Constraints

Project: Library Loan Management System

Database: MySQL 8.0

Objective

Explain and verify primary keys, foreign keys, UNIQUE, CHECK, NOT NULL, DEFAULT, and the rule that each copy can have only one open loan.

Note: outputs are expected results based on the included sample data. Run the SQL in your MySQL environment and insert your own screenshots if required by your instructor.

1. Introduction

Integrity constraints keep data valid and prevent broken relationships. The constraints are defined in the Assignment 5 schema and tested below.

2. Constraints Used

PRIMARY KEY: identifies each member, book, author, copy, and loan; BookAuthors uses a composite key.
FOREIGN KEY: ensures each author-book link, copy, and loan refers to an existing record.
UNIQUE: prevents duplicate emails, accession numbers, duplicate author links, and multiple open loans for the same copy.
NOT NULL: requires essential names, emails, titles, categories, dates, and relationship IDs.
CHECK: restricts membership status, copy condition, and valid loan date order.
DEFAULT: sets MembershipStatus to Active and ConditionStatus to Good when omitted.

3. SQL Schema

```
Run the complete DDL from Assignment 5 before these tests. The key definition for preventing two open
loans is:
ALTER TABLE Loans ADD COLUMN OpenCopyID INT GENERATED ALWAYS AS (CASE WHEN ReturnedOn IS NULL THEN
CopyID ELSE NULL END) STORED;
ALTER TABLE Loans ADD CONSTRAINT uq_one_open_loan_per_copy UNIQUE (OpenCopyID);

Note: execute these ALTER statements only if OpenCopyID and the constraint are not already present; the
Assignment 5 CREATE TABLE script already includes them.
```

4. Verification Queries

```
-- Verify the default membership status
INSERT INTO Members (MemberID, FullName, Email, JoinedOn)
VALUES (990, 'Test Member', 'test990@example.com', '2026-10-22');
SELECT MemberID, MembershipStatus FROM Members WHERE MemberID = 990;

-- Invalid membership status: should fail CHECK
INSERT INTO Members (MemberID, FullName, Email, JoinedOn, MembershipStatus)
VALUES (991, 'Invalid Status', 'invalid991@example.com', '2026-10-22', 'Paused');

-- Invalid loan date order: should fail CHECK
INSERT INTO Loans (LoanID, MemberID, CopyID, BorrowedOn, DueOn)
VALUES (9990, 990, 516, '2026-10-25', '2026-10-20');

-- Duplicate accession number: should fail UNIQUE
INSERT INTO BookCopies (CopyID, BookID, AccessionNumber, ShelfLocation)
VALUES (990, 108, 'LIB-016', 'Z9-S9');

-- Inspect constraints
SHOW CREATE TABLE Members;
SHOW CREATE TABLE BookCopies;
SHOW CREATE TABLE Loans;
```

5. Expected Results

The valid test member should be inserted with MembershipStatus = Active. The invalid status, invalid due-date order, and duplicate accession number should be rejected. Run each negative test separately because a failed statement may interrupt a script depending on the SQL client. Remove the valid test row after testing, provided no loan references it.

6. Open Loan Integrity Test

The generated OpenCopyID is equal to CopyID only while ReturnedOn is NULL. For returned loans it is NULL, and MySQL permits multiple NULL values in a UNIQUE index. This allows a copy to have many historical loans but only one open loan at a time. The sample data demonstrates that loan 9000 is returned and loan 9001 is the current open loan for copy 501.

7. Compatibility Note

CHECK constraints are enforced in MySQL 8.0.16 and later. Generated columns and their syntax depend on the DBMS version. Confirm the version and test constraints in your own environment.

8. Conclusion

The constraints maintain entity identity, valid references, allowed values, and loan availability integrity. Verification should include both valid inserts and invalid data that must be rejected.